

User Guide

Integrating Amazon S3 Tables with Amazon QuickSight (Quick Suite)

Amazon Web Services

April 2026

Introduction

This guide walks you through the step-by-step process of integrating Amazon S3 Tables with Amazon QuickSight (Quick Suite) to create data analyses. S3 Tables provide a managed Apache Iceberg table experience on S3, and QuickSight can connect directly to these tables as a data source for visualization and analysis.

Prerequisites

- An AWS account with appropriate permissions
- Access to Amazon QuickSight (Quick Suite)
- An existing S3 Table bucket with table data
- The ARN of your S3 Table

Workflow Overview

1. Manage your AWS account in QuickSight
2. Enable S3 Tables as an allowed AWS resource
3. Create a new data source using the S3 Table ARN
4. Create a dataset from the data source
5. Generate an analysis from the dataset

Step 1: Access Account Management

Begin by navigating to the QuickSight console. Click on the user icon or account menu in the top-right corner to access the Manage QuickSight option.

1.1 Open the QuickSight Account Menu

From the QuickSight home page, click on the profile icon in the top-right corner. This opens the account management dropdown where you can access subscription and security settings.

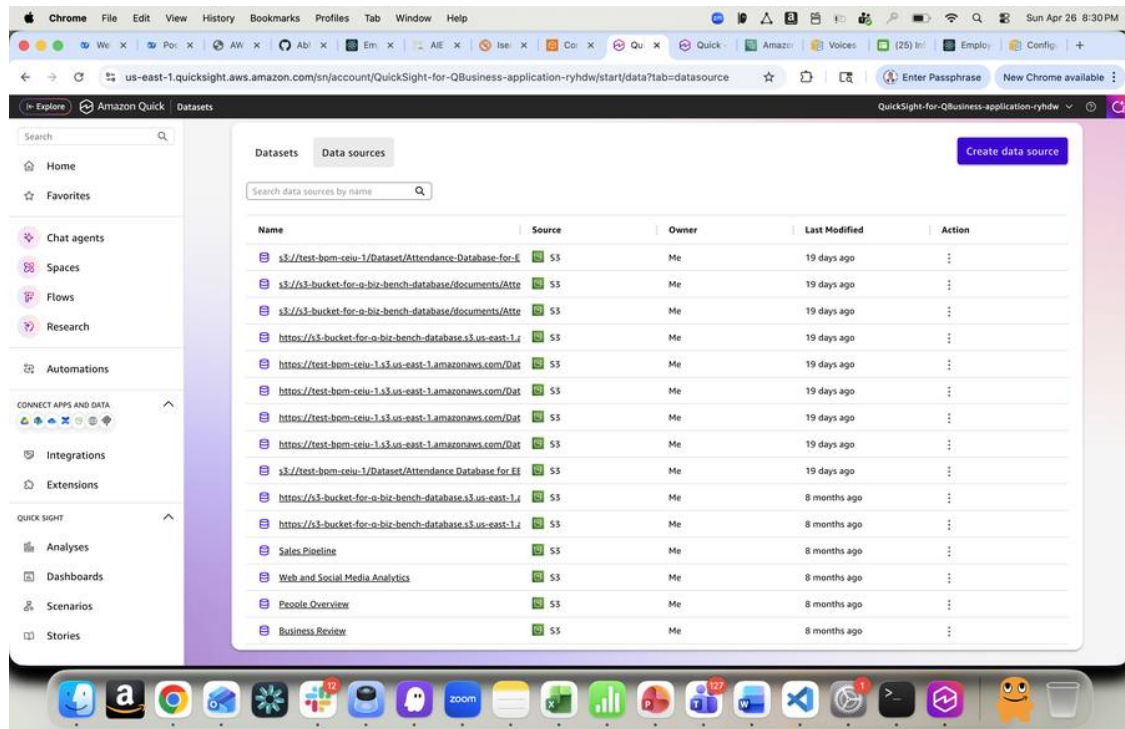


Figure 1: QuickSight home page — click the user/account icon in the top-right

1.2 Navigate to Manage QuickSight

Select "Manage QuickSight" from the dropdown menu. This takes you to the account administration panel.

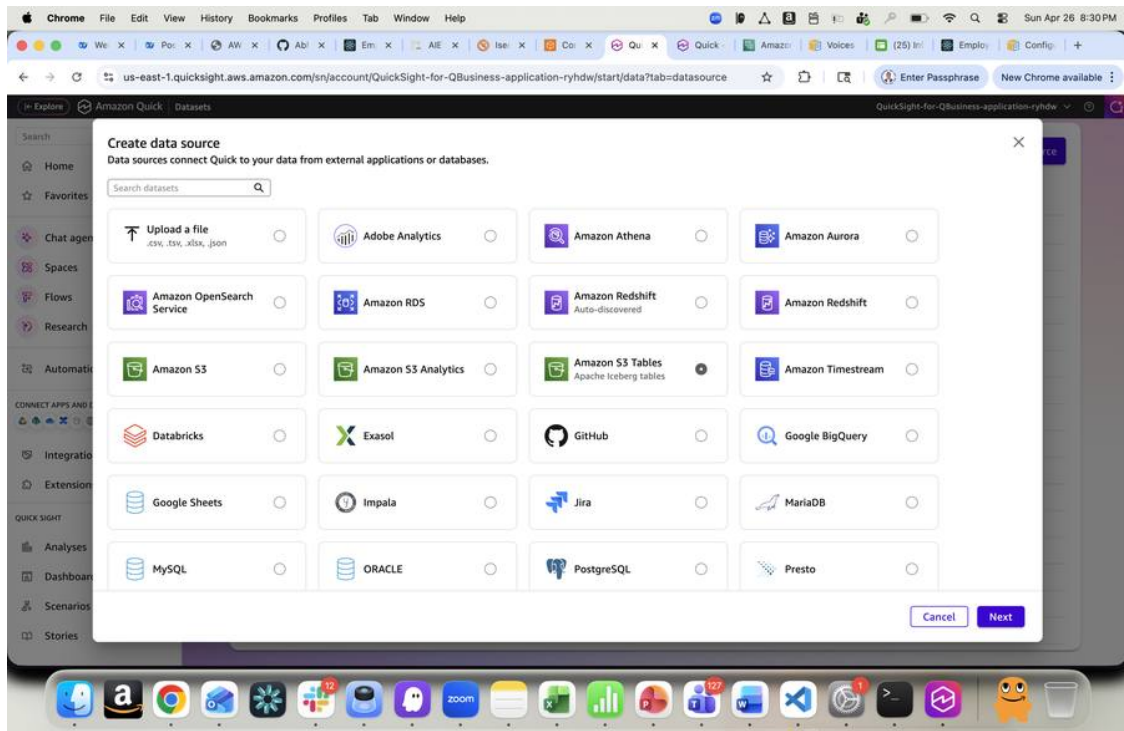


Figure 2: Select "Manage QuickSight" from the account menu

2.1 Access Security & Permissions

The screenshot displays the Amazon QuickSight console interface. A modal dialog titled "New Amazon S3 Tables data source" is open in the center. The dialog contains two input fields: "Data source name" with a placeholder "Enter a name for the data source" and "S3 table bucket ARN" with a placeholder "Enter the table bucket ARN to use". A "Create data source" button is located at the bottom right of the dialog. In the background, the "Datasets" tab is active, showing a table of existing data sources. The table has columns for "Name", "Last Modified", and "Action". The data sources listed include various S3 buckets (e.g., s3://test-bom-1, s3://s3-bucket-1) and Amazon Redshift instances (e.g., https://s3-bucket-for-a-biz-bench-database.s3.us-east-1.amazonaws.com/Attendance Database for E). The top of the screen shows the Chrome browser tabs and the system clock indicating Sun Apr 26 8:31 PM.

2.2 Manage QuickSight Access to AWS Services

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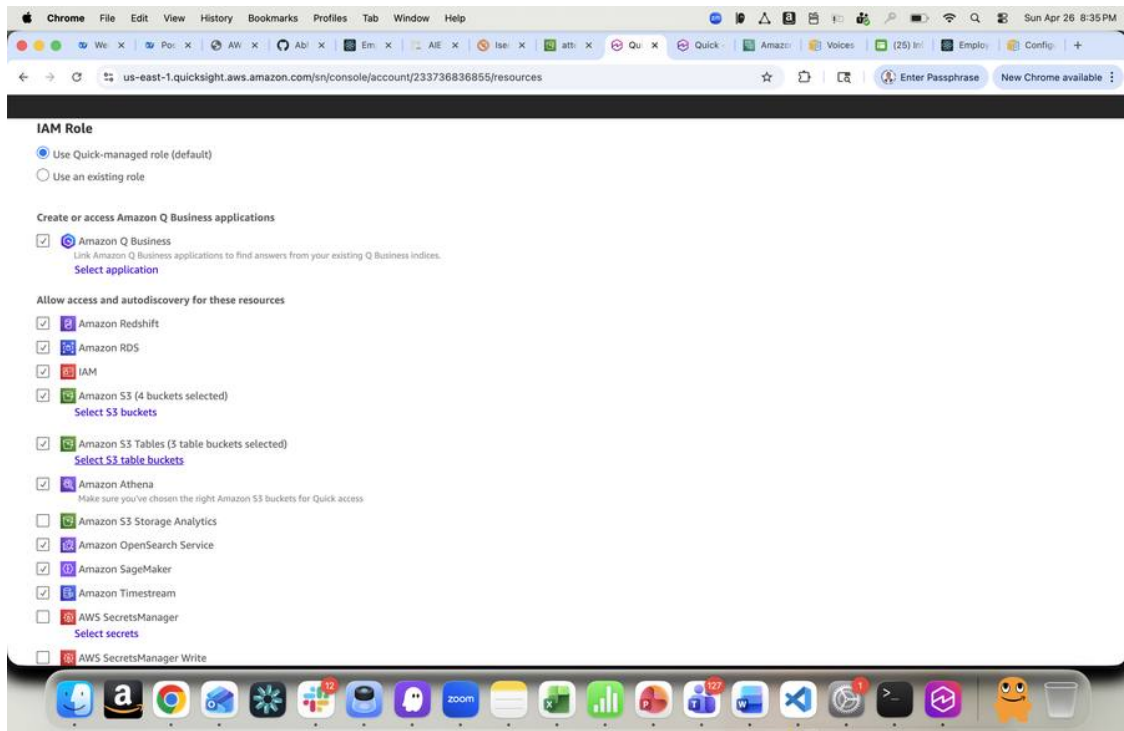


Figure 4: Security & permissions — click "Manage" under AWS services access

Note: You must have QuickSight admin privileges to modify these security settings. Contact your account administrator if you do not see the Manage option.

Step 3: Enable S3 Tables Access in QuickSight

Now you need to explicitly allow QuickSight to access Amazon S3 Tables. This involves enabling the S3 Tables option in the AWS resource permissions panel.

3.1 Enable Amazon S3 Tables

In the AWS services access configuration panel, locate the "Amazon S3 Tables" option. Check the box to allow QuickSight to access S3 Table resources. This grants the IAM role used by QuickSight the necessary permissions to read data from S3 Tables.

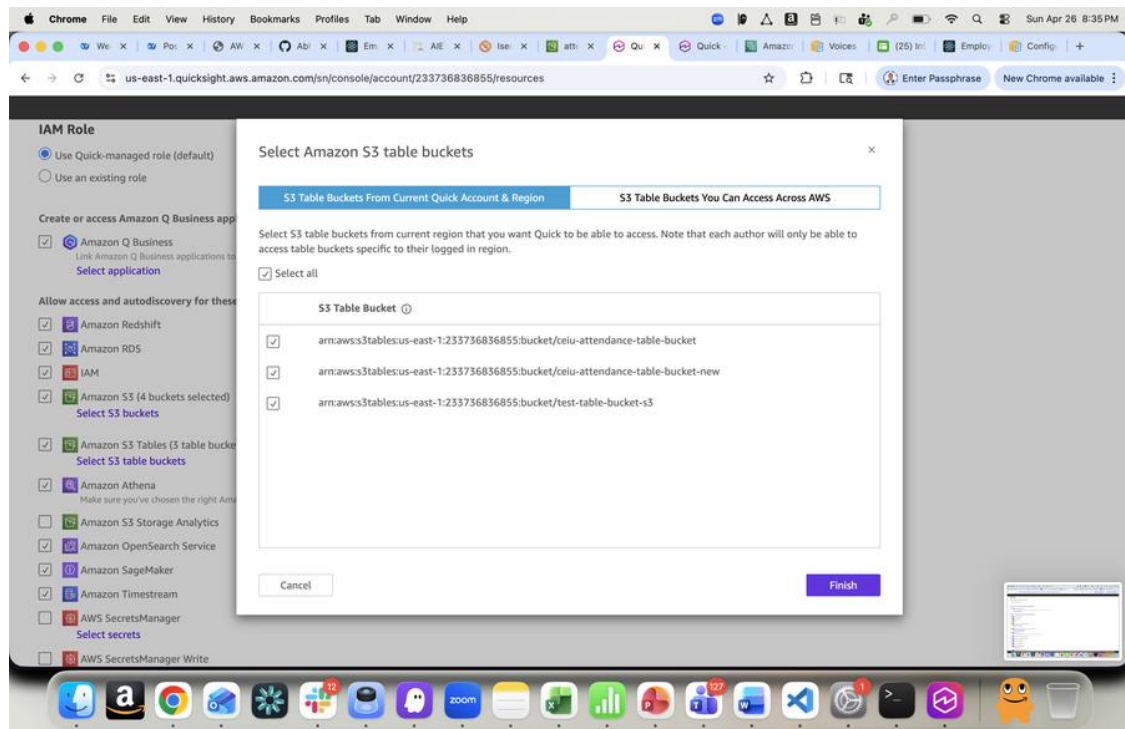


Figure 5: Enable Amazon S3 Tables in the AWS resource permissions

3.2 Confirm S3 Tables Permission

Verify that Amazon S3 Tables is now checked/enabled in the list. You may also see other AWS services listed here. Ensure the S3 Tables toggle is turned on, then save your changes.

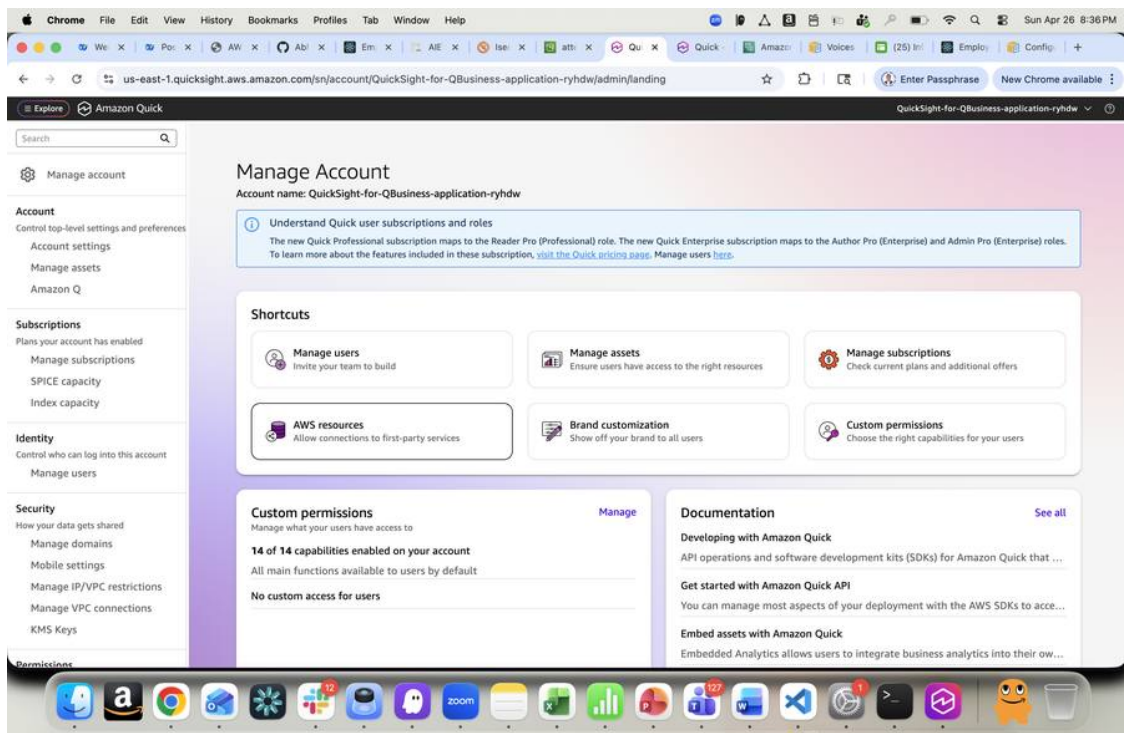


Figure 6: S3 Tables permission enabled — confirm and save

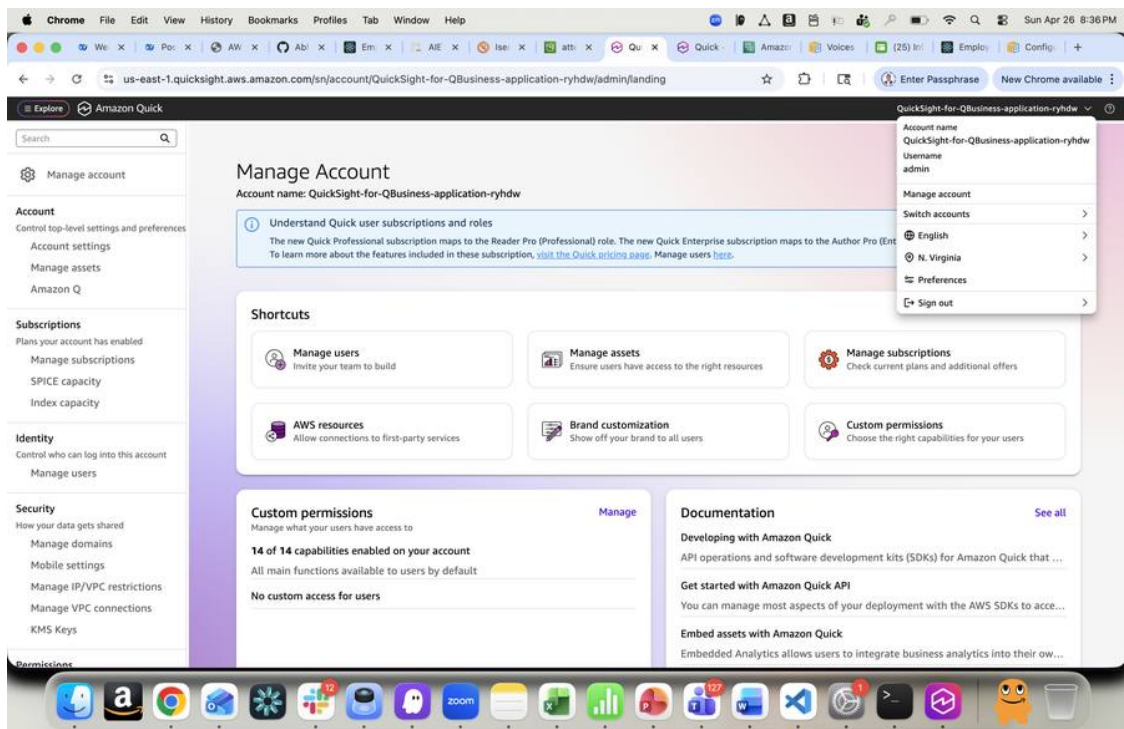


Figure 7: Updated permissions showing S3 Tables access is now active

Note: The IAM role associated with QuickSight must have the necessary policies to access S3 Table buckets. If you encounter permission errors later, verify the IAM role policies in the AWS IAM console.

Step 4: Create a Data Source Using S3 Tables

With S3 Tables access enabled, you can now create a new data source in QuickSight that connects to your S3 Table. This is done from the Datasets section.

4.1 Navigate to Datasets

From the QuickSight home page, click on "Datasets" in the left navigation panel. This takes you to the datasets management page where you can create new datasets and data sources.

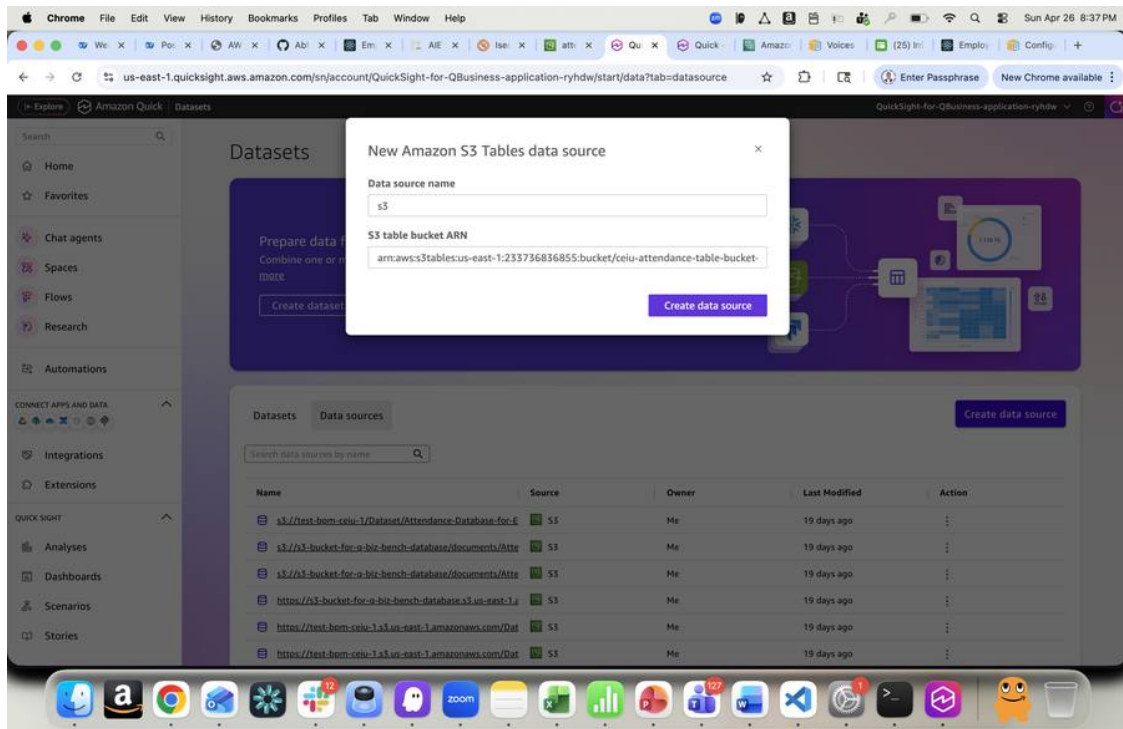


Figure 8: Navigate to the Datasets section from the left navigation

4.2 Create a New Dataset

Click the "New dataset" button in the top-right corner to begin the process of creating a new data source connection.

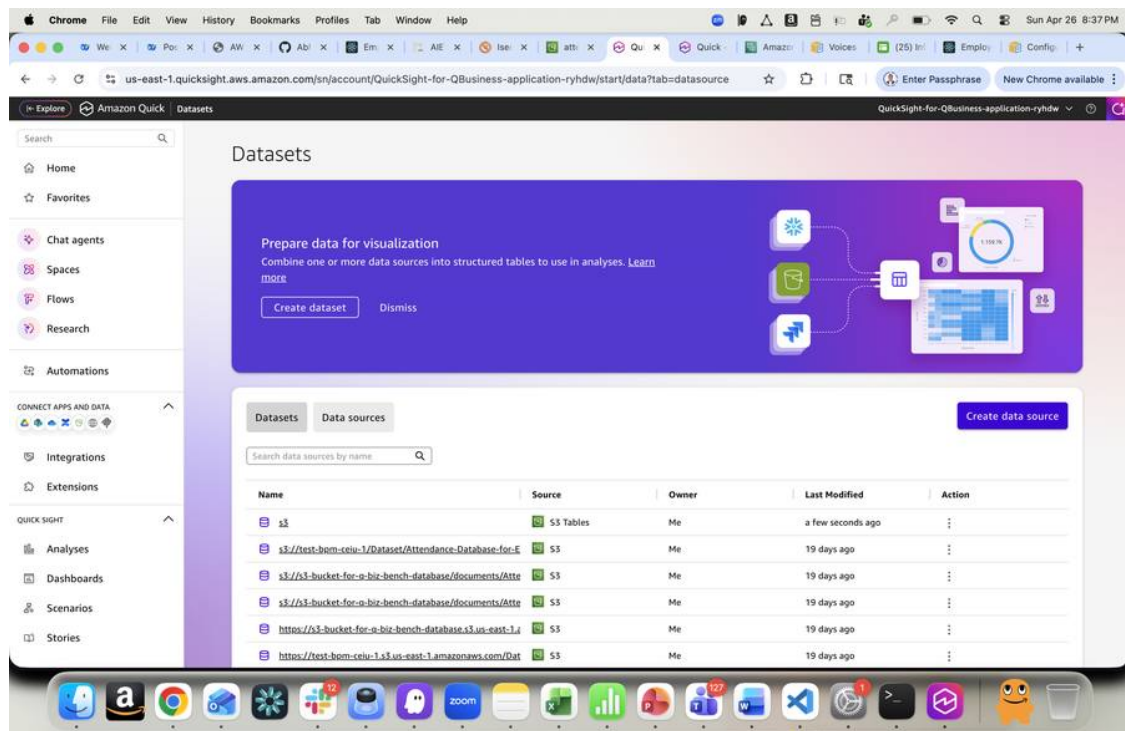


Figure 9: Click "New dataset" to start the data source creation

4.3 Select S3 Tables as Data Source

In the data source selection page, locate and select "Amazon S3 Tables" from the available connectors. This option appears because you enabled S3 Tables access in the previous step.

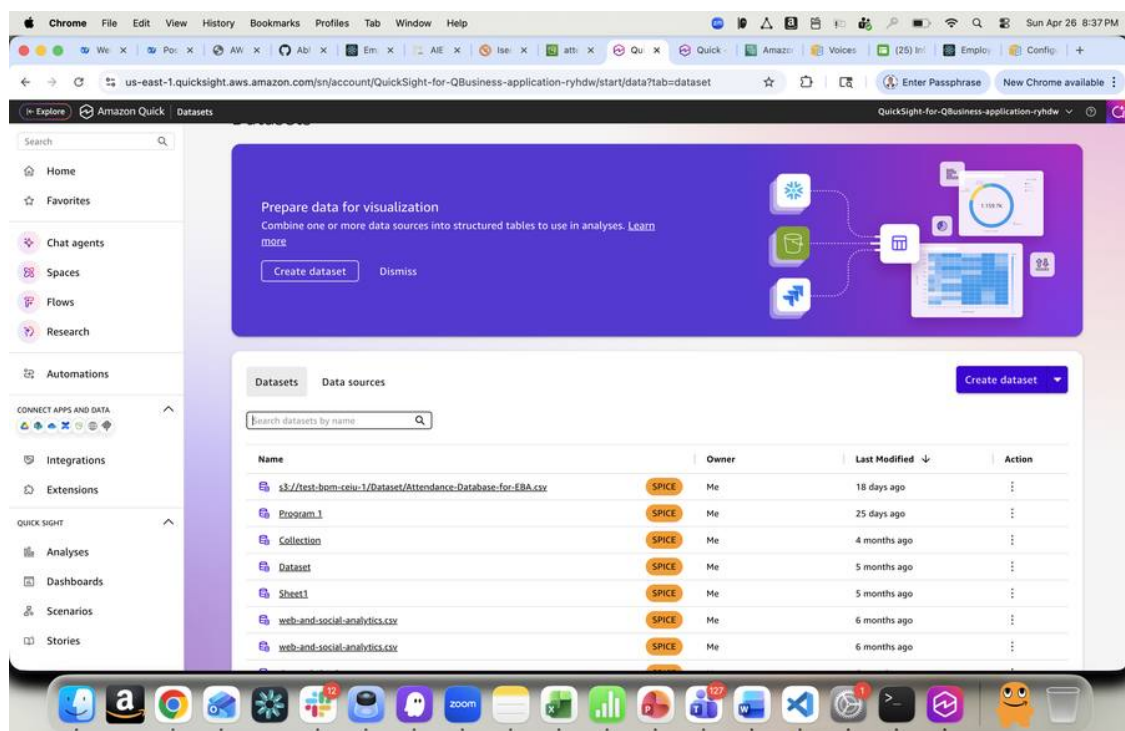


Figure 10: Select "Amazon S3 Tables" from the data source connector list

4.4 Configure the S3 Table Connection

Enter a descriptive name for the data source. You will need the ARN (Amazon Resource Name) of your S3 Table bucket to establish the connection.

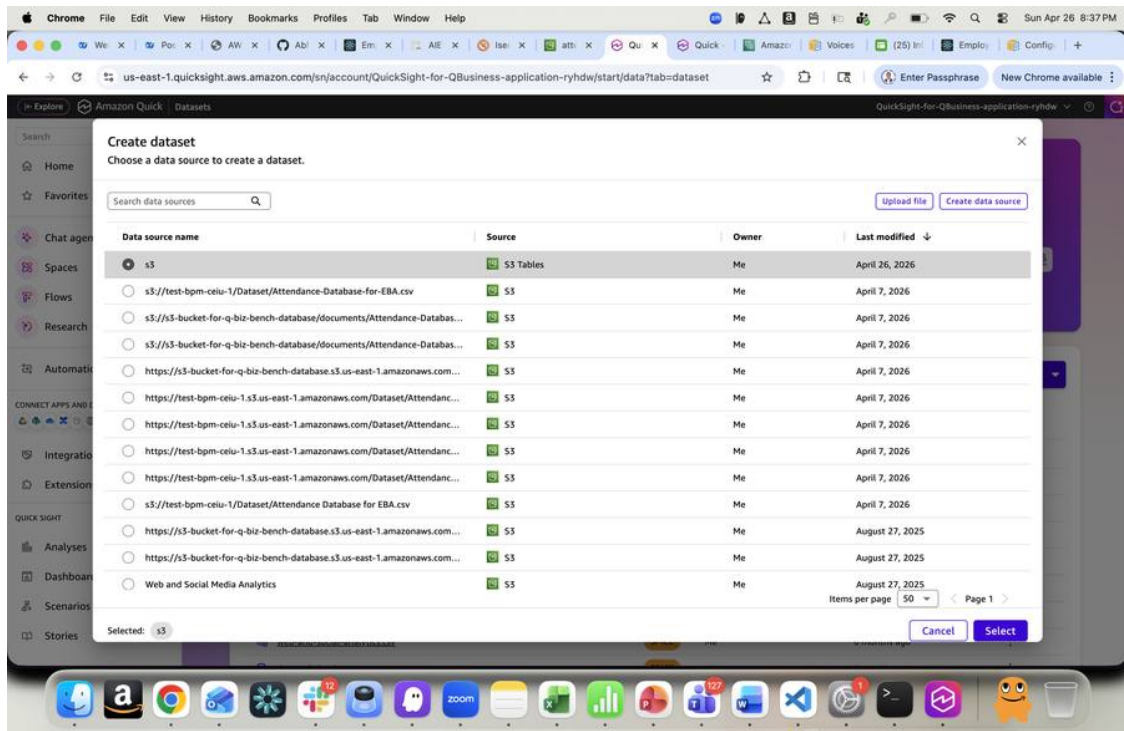


Figure 11: Configure the S3 Table data source — enter a name

Step 5: Provide the S3 Table ARN

You need the ARN of your S3 Table to connect QuickSight. You can find this ARN in the Amazon S3 console under the Tables section.

5.1 Locate the S3 Table ARN

Open the Amazon S3 console in a separate tab. Navigate to the S3 Tables section and find your table bucket. Copy the Table bucket ARN from the properties.

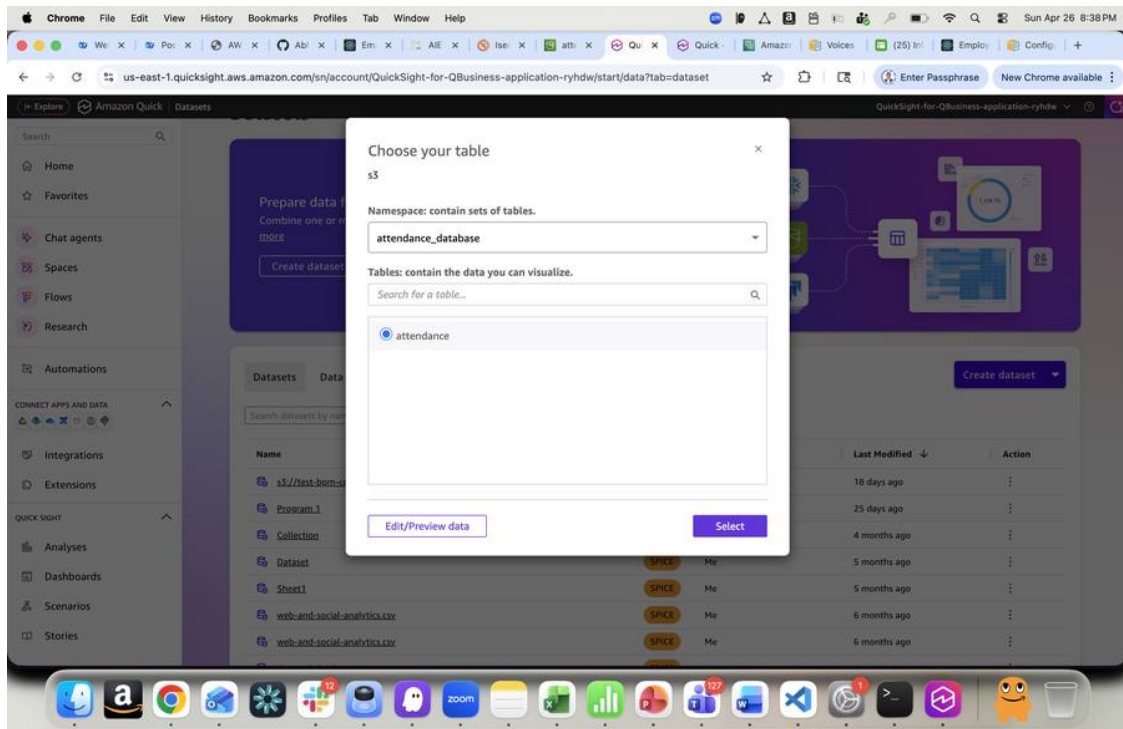


Figure 12: Locate your S3 Table ARN in the S3 console

5.2 Enter the ARN in QuickSight

Return to the QuickSight data source configuration page and paste the S3 Table bucket ARN into the ARN field. This connects QuickSight to your specific S3 Table resource.

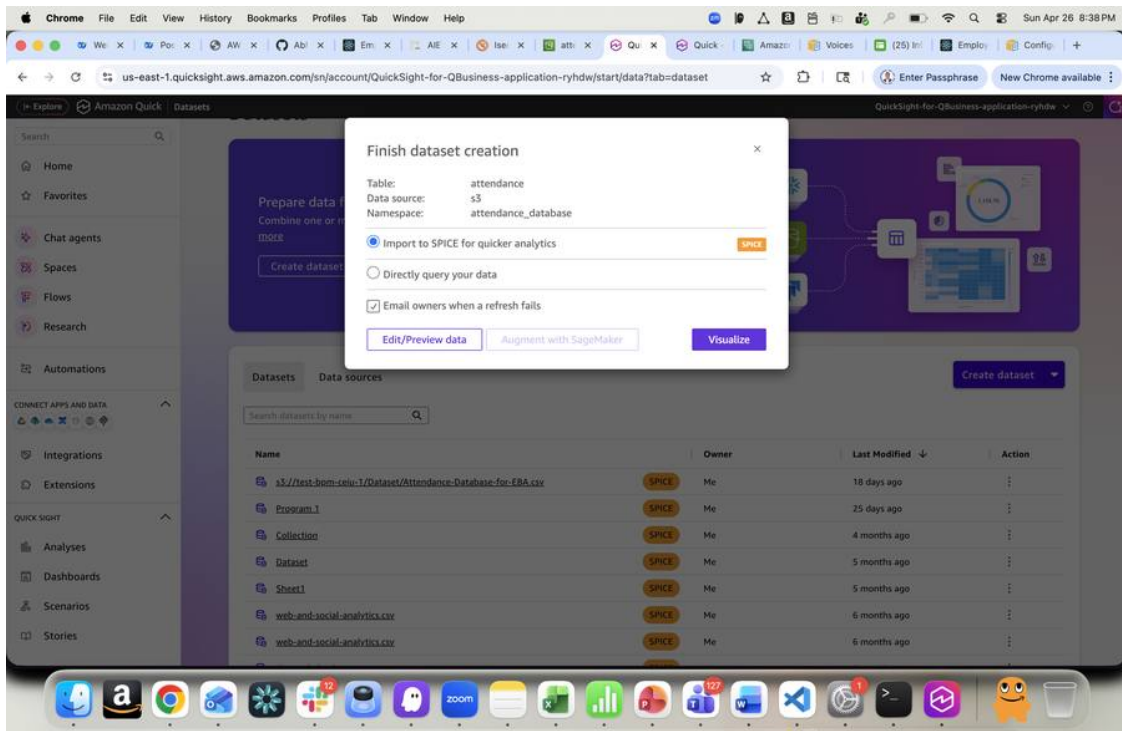


Figure 13: Paste the S3 Table bucket ARN into the data source configuration

5.3 Validate and Create the Data Source

Click "Validate connection" to verify that QuickSight can access the S3 Table. Once validated, click "Create data source" to finalize the connection.

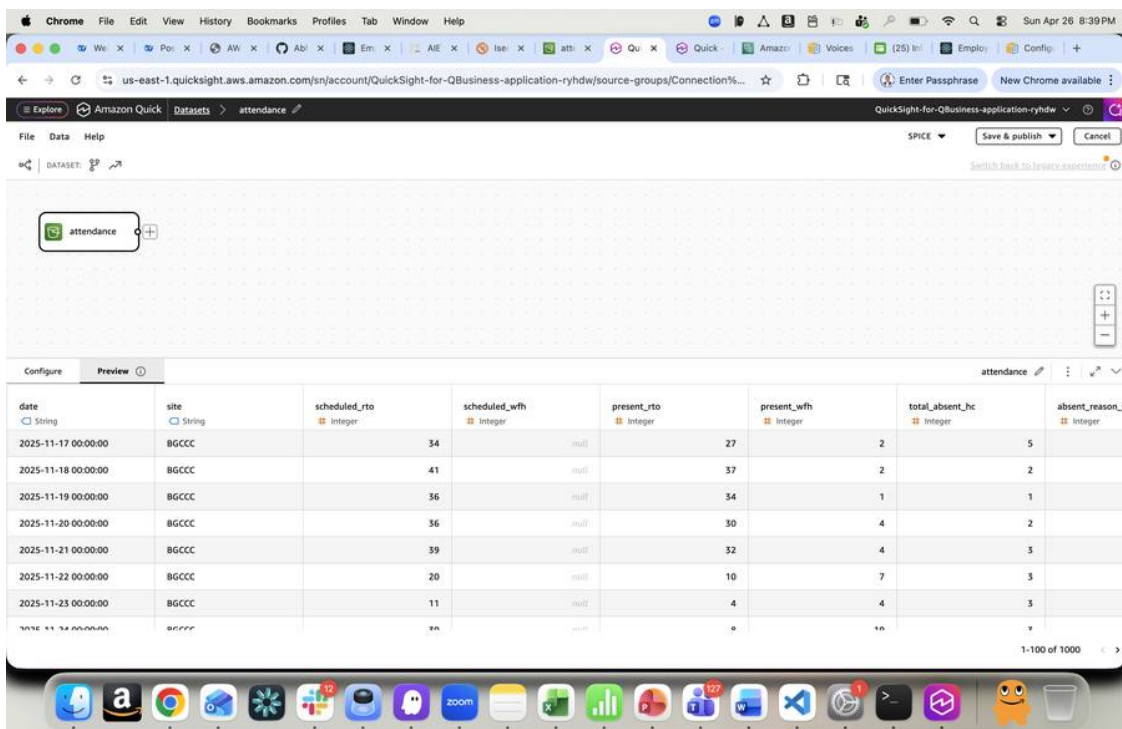


Figure 14: Validate connection and create the data source

Note: If the connection validation fails, verify: (1) The S3 Table ARN is correct, (2) The QuickSight IAM role has the required S3 Tables permissions, and (3) S3 Tables is enabled in the Security & Permissions settings.

Step 6: Create a Dataset from the Data Source

After creating the data source, you need to create a dataset. A dataset defines which tables and columns from the data source will be available for analysis in QuickSight.

6.1 Select the Table from the Data Source

After the data source is created, QuickSight will display the available tables (namespaces) from your S3 Table bucket. Select the table you want to analyze.

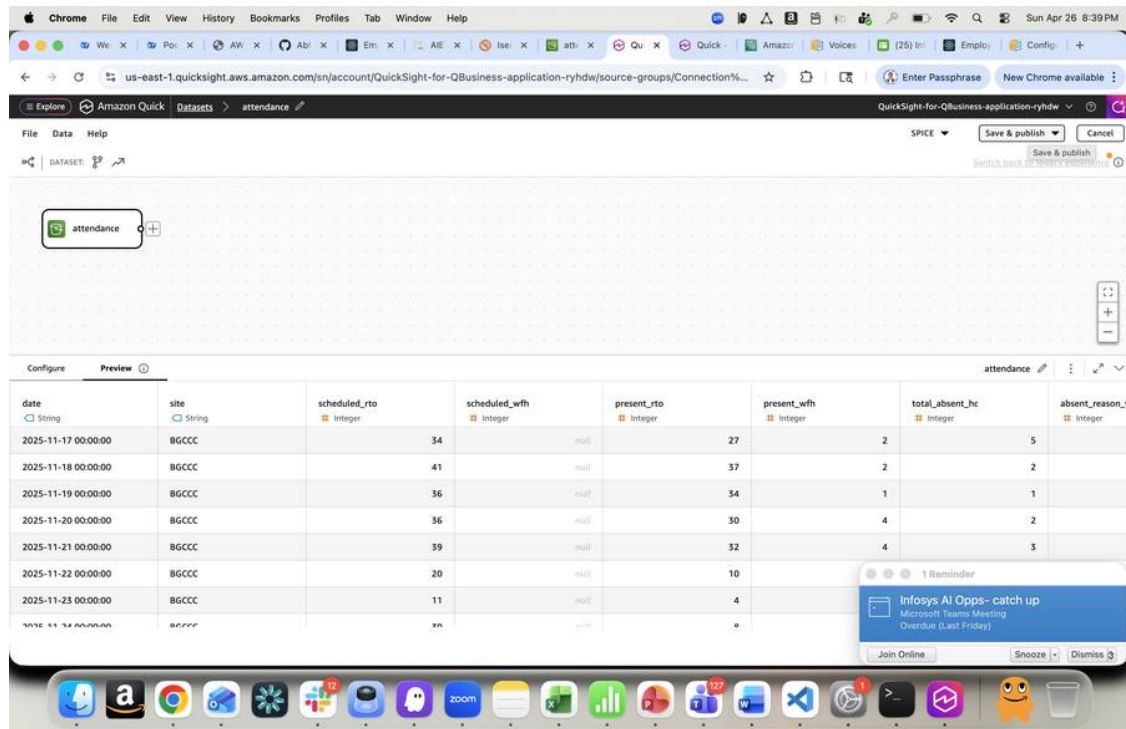


Figure 15: Select the table from the S3 Table data source

6.2 Choose the Database/Namespace

Select the appropriate database or namespace from the dropdown. S3 Tables organize data in namespaces, and you need to select the one that contains your target table.

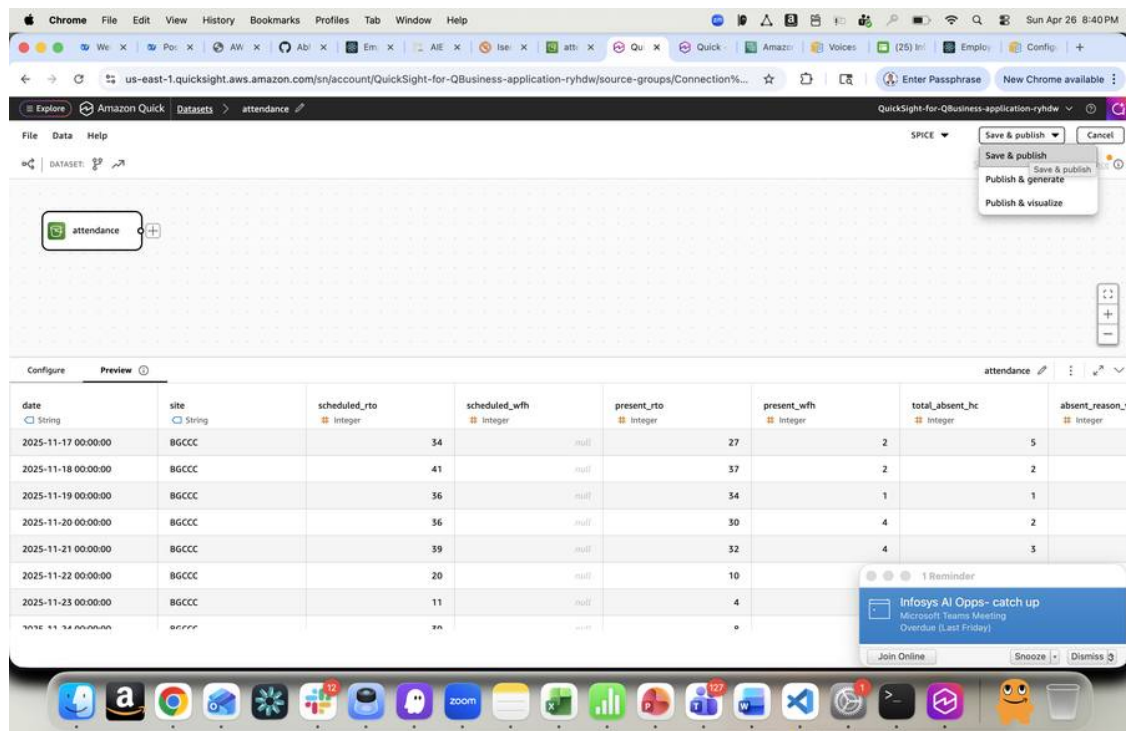


Figure 16: Choose the database/namespace for your dataset

6.3 Select the Specific Table

Browse the available tables within the selected namespace and choose the specific table you want to bring into QuickSight for analysis.

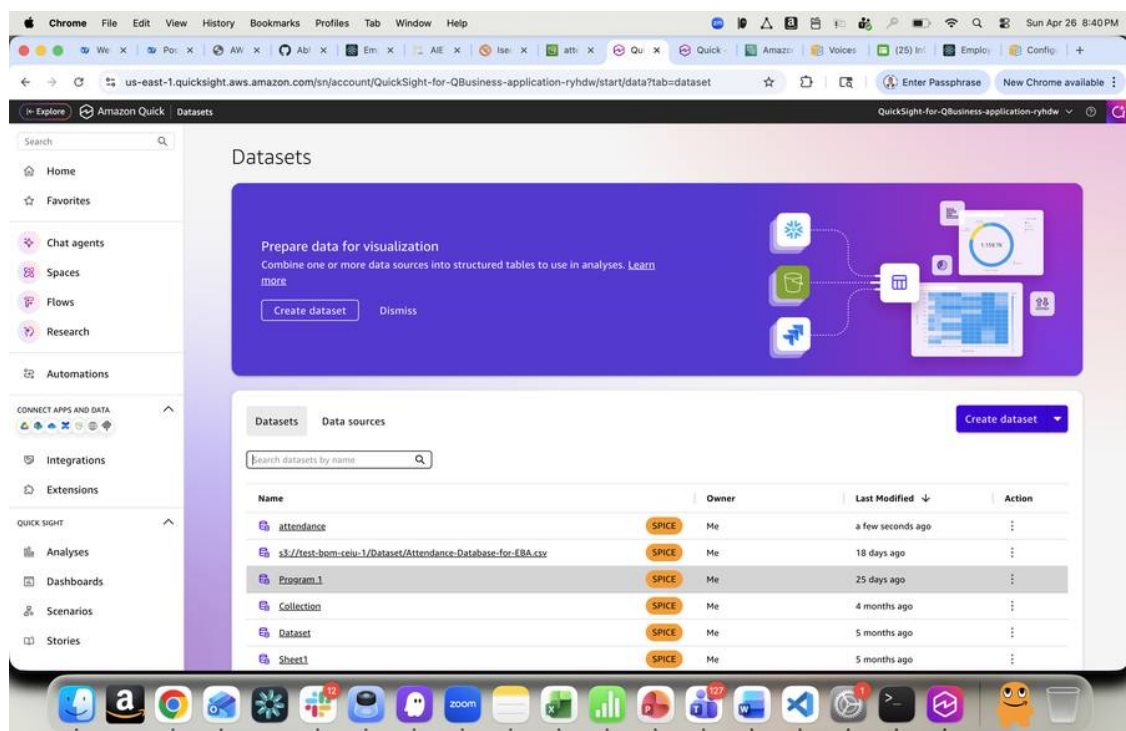


Figure 17: Select the specific table within the namespace

6.4 Configure Dataset Settings

Review the dataset configuration options. You can choose to import the data into SPICE (Super-fast, Parallel, In-memory Calculation Engine) for faster performance, or query the data directly.

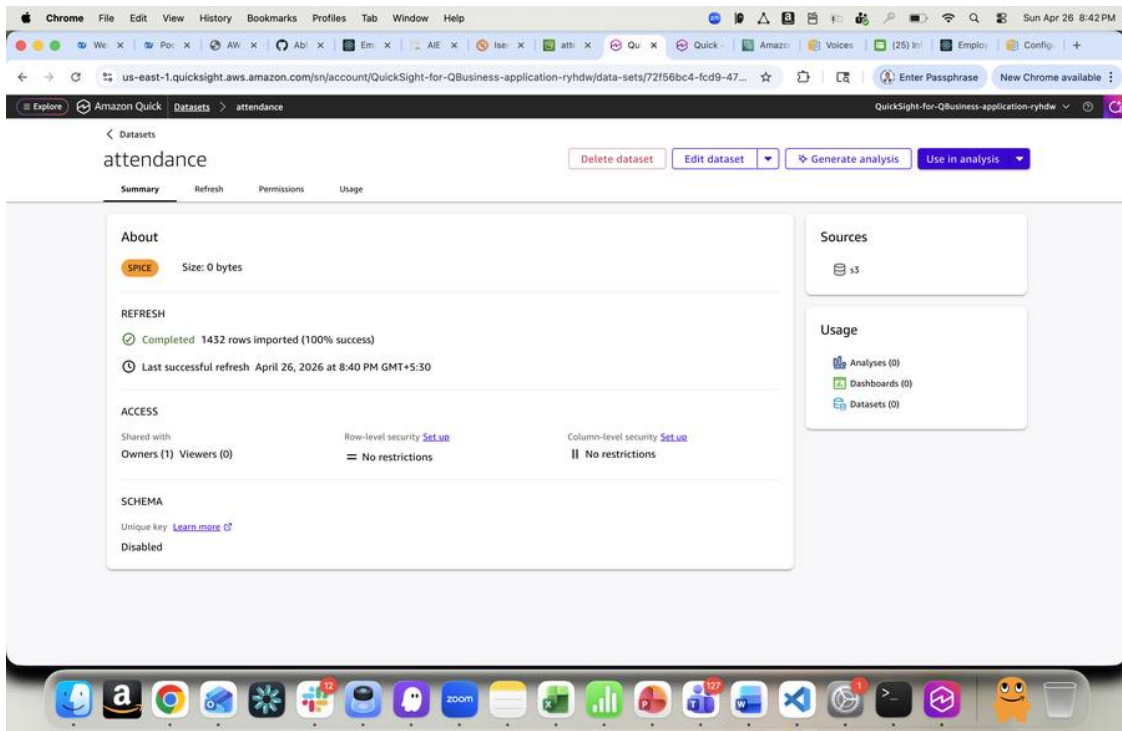


Figure 18: Dataset configuration — choose import to SPICE or direct query

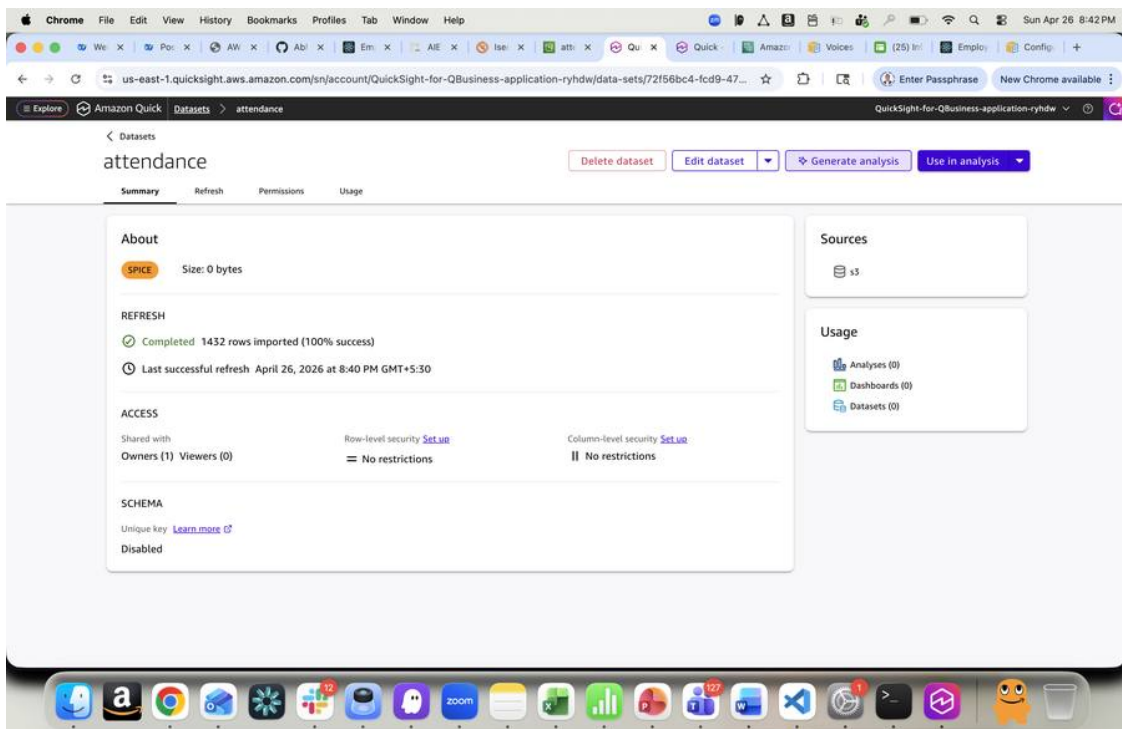


Figure 19: Finalize dataset creation and review the schema

Note: SPICE provides faster query performance by caching data in memory. For large datasets, consider SPICE import. For data that changes frequently, direct query ensures you always see the latest data.

Step 7: Generate an Analysis

With the dataset created, you can now create a QuickSight analysis to visualize and explore your S3 Table data. QuickSight offers AI-powered auto-generation of analyses.

7.1 Create a New Analysis

From the dataset page, click "Use in Analysis" or navigate to Analyses and create a new one. Select the dataset you just created as the data source for the analysis.

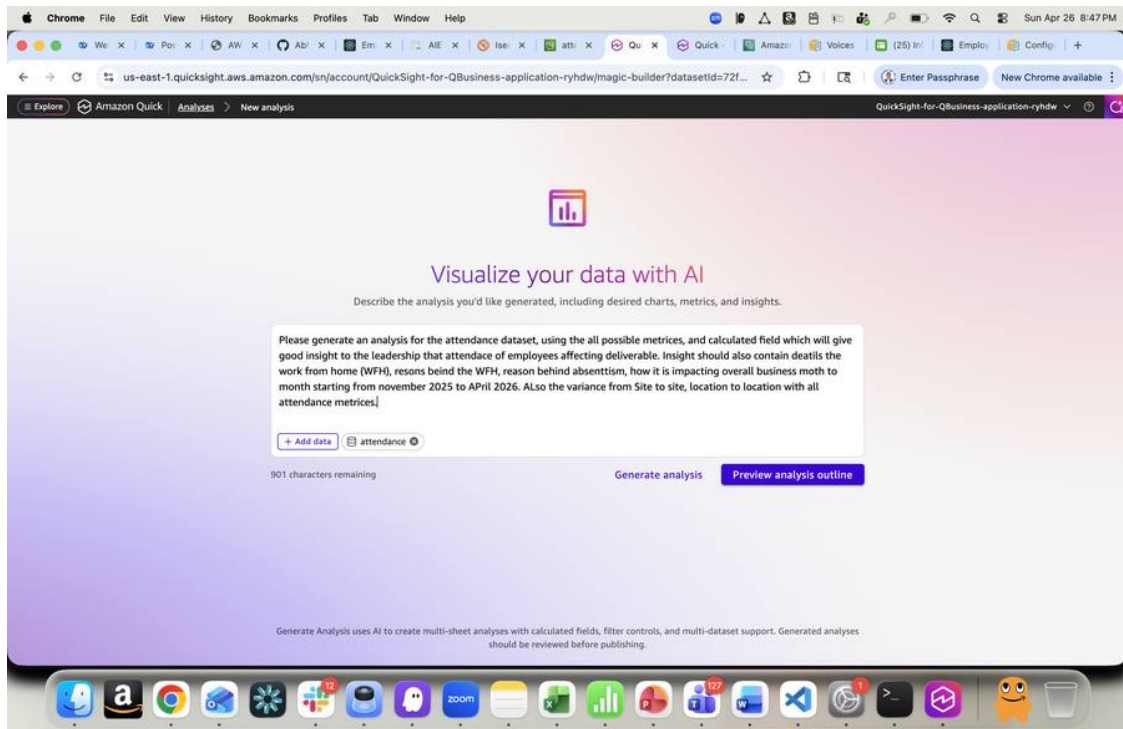


Figure 20: Create a new analysis using the S3 Table dataset

7.2 Use Generate Analysis (AI-Powered)

QuickSight offers a "Generate Analysis" feature powered by AI. Click on "Generate" to let QuickSight automatically create visualizations and insights based on your data structure and content.

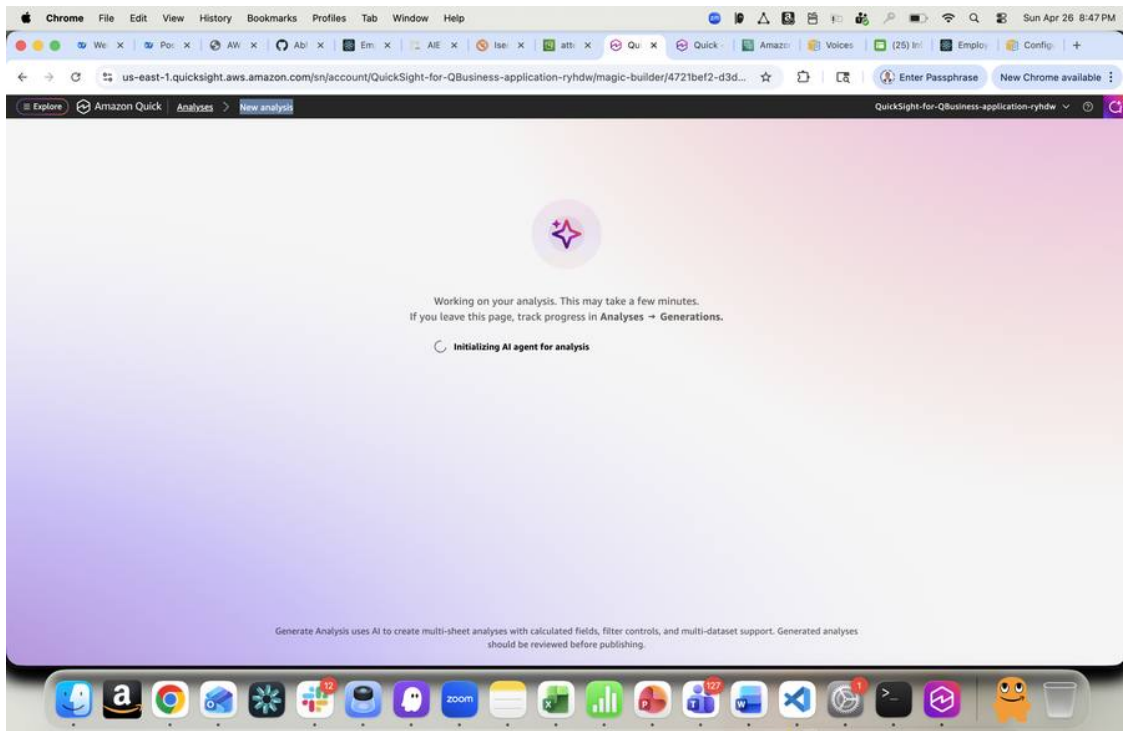


Figure 21: Use the "Generate Analysis" feature for AI-powered insights

7.3 Review the Generated Analysis

QuickSight generates a complete analysis with multiple visualizations — including charts, tables, and KPIs. Review the generated sheets and customize them as needed.

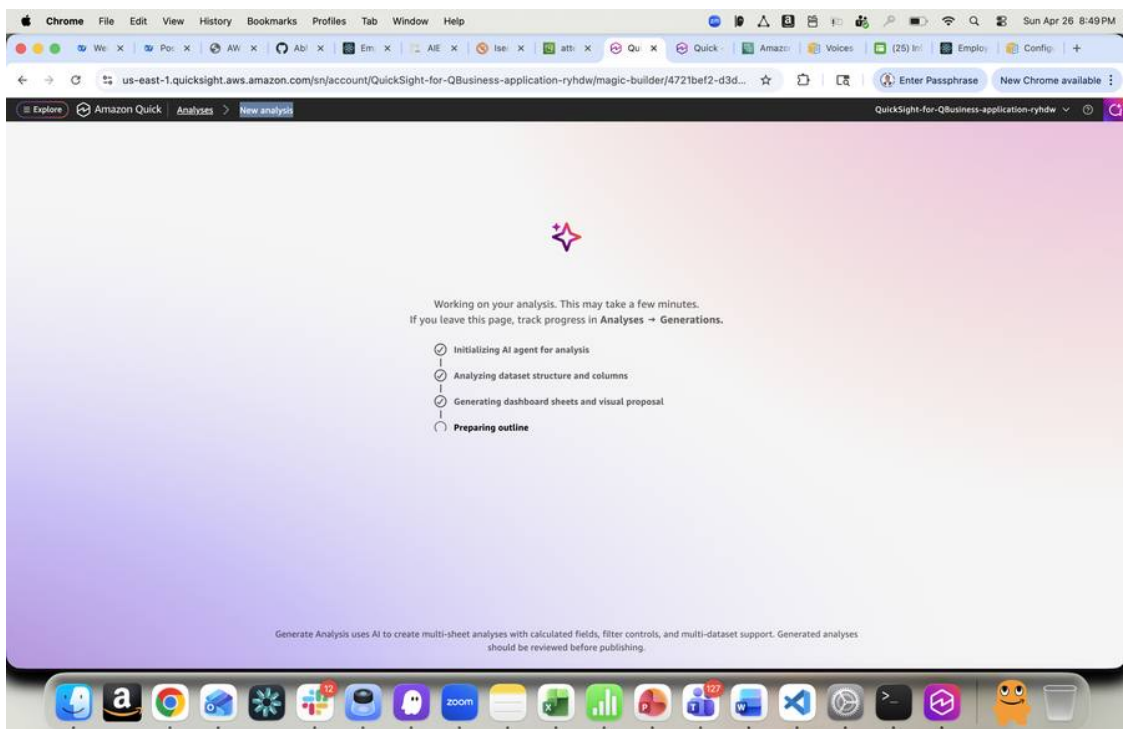


Figure 22: AI-generated analysis with visualizations from S3 Table data

7.4 Explore and Customize

Explore the generated analysis. You can add new visual types, apply filters, create calculated fields, and customize the layout. The analysis is now connected live to your S3 Table data.

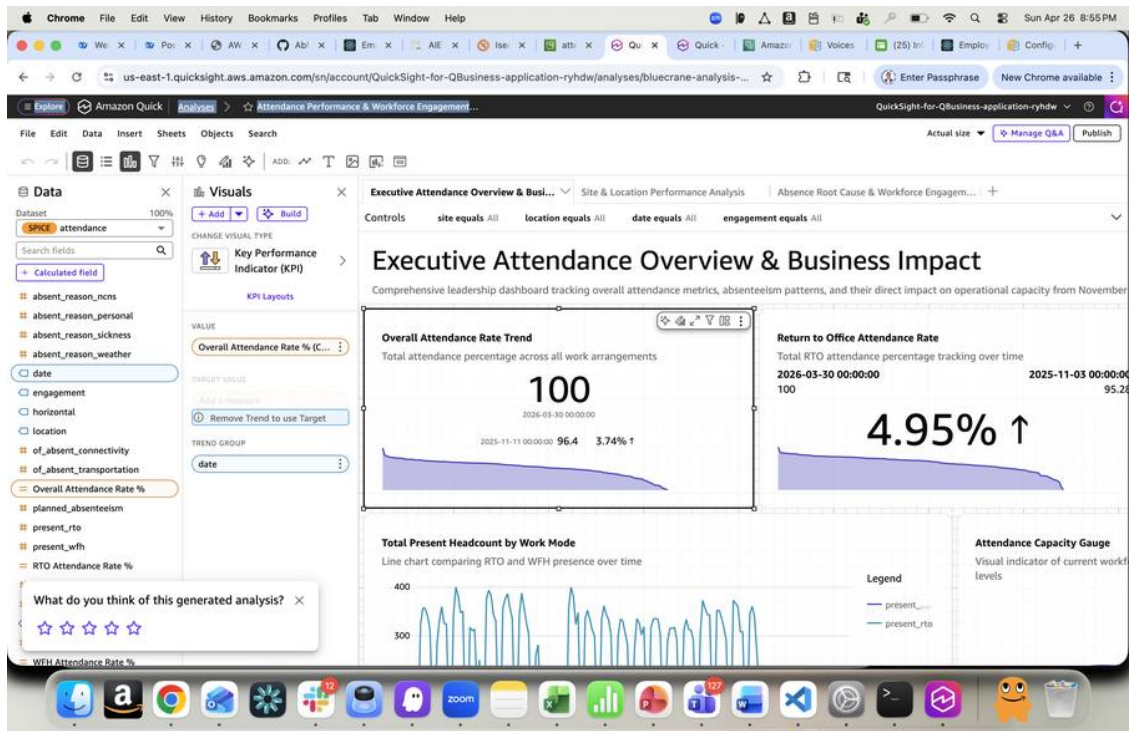


Figure 23: Final analysis — fully interactive with S3 Table data

Summary

You have successfully integrated Amazon S3 Tables with Amazon QuickSight. Here is a quick recap of the steps covered:

Step	Action	Where
1	Access Account Management	QuickSight > Account menu > Manage QuickSight
2	Manage AWS Resource Permissions	Manage QuickSight > Security & permissions
3	Enable S3 Tables Access	Security & permissions > AWS services > S3 Tables
4	Create Data Source	Datasets > New dataset > Amazon S3 Tables
5	Provide S3 Table ARN	Data source config > ARN field (from S3 console)
6	Create Dataset	Select namespace > Select table > SPICE or Direct
7	Generate Analysis	Analyses > Generate Analysis (AI-powered)

For further assistance, refer to the AWS documentation for Amazon S3 Tables and Amazon QuickSight, or contact your AWS Solutions Architect.